

Agent Hive

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QpiAI: Profile Snapshot

❑ Profile

- **100+** strong **Engineering, R&D and Product** team
- **HQ** in **Bengaluru** with presence in **Finland** and **USA**
- **40+** patents filed

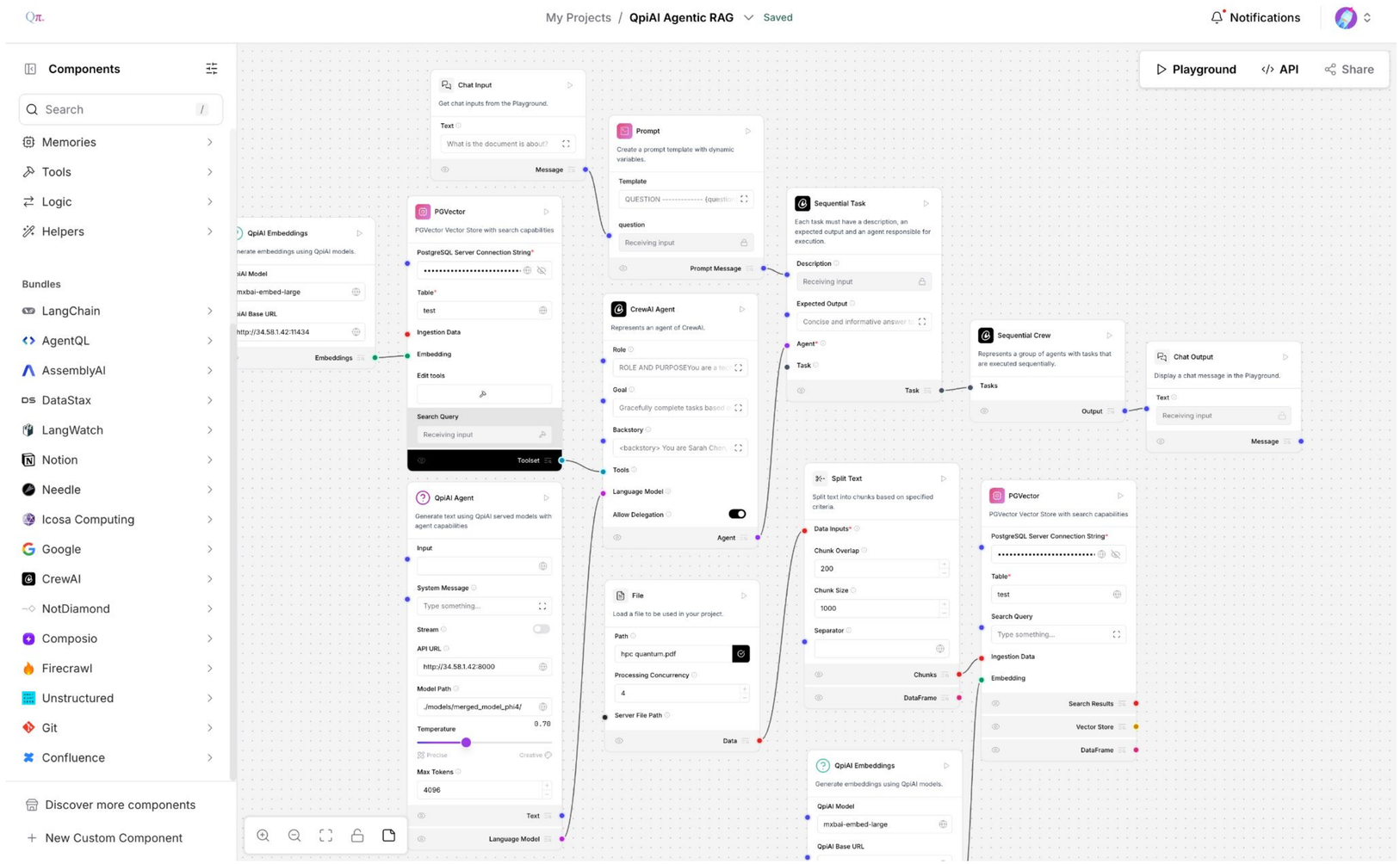
❑ Recognitions

- **Future AI Pioneer** at **Asia Pacific Conference of German Business** - APK 2024
- **NASSCOM AI Gamechanger of the Year 2024** in Manufacturing Category
- Mint Digital Innovation Summit 2023 - **Winner for AI Category**
- DoT Digital Communication Innovation **Scheme 2023 Winner**
- Accenture TechNext Advanced Computing Challenge 2023 **Finalist**
- **NASSCOM Emerge 50** and **Enterprise Product of the Year Award 2021**
- Infosys Living Labs Partner and TCS Co-Innovation Partner

❑ Featured in

QpiAI Agent Hive

- GUI based Multi Agent Systems, Agentic RAG and SLM/LLM Fine Tuning Platform.
- Swarm of specialist SLMs based **Domain Targeted Agents** to drive accuracy and cost efficiency over Frontier LLMs based agents. Drive **10x to 50x cost reduction** for complex use cases.
- **10x Productivity Gain** with Comprehensive Tools Integration.

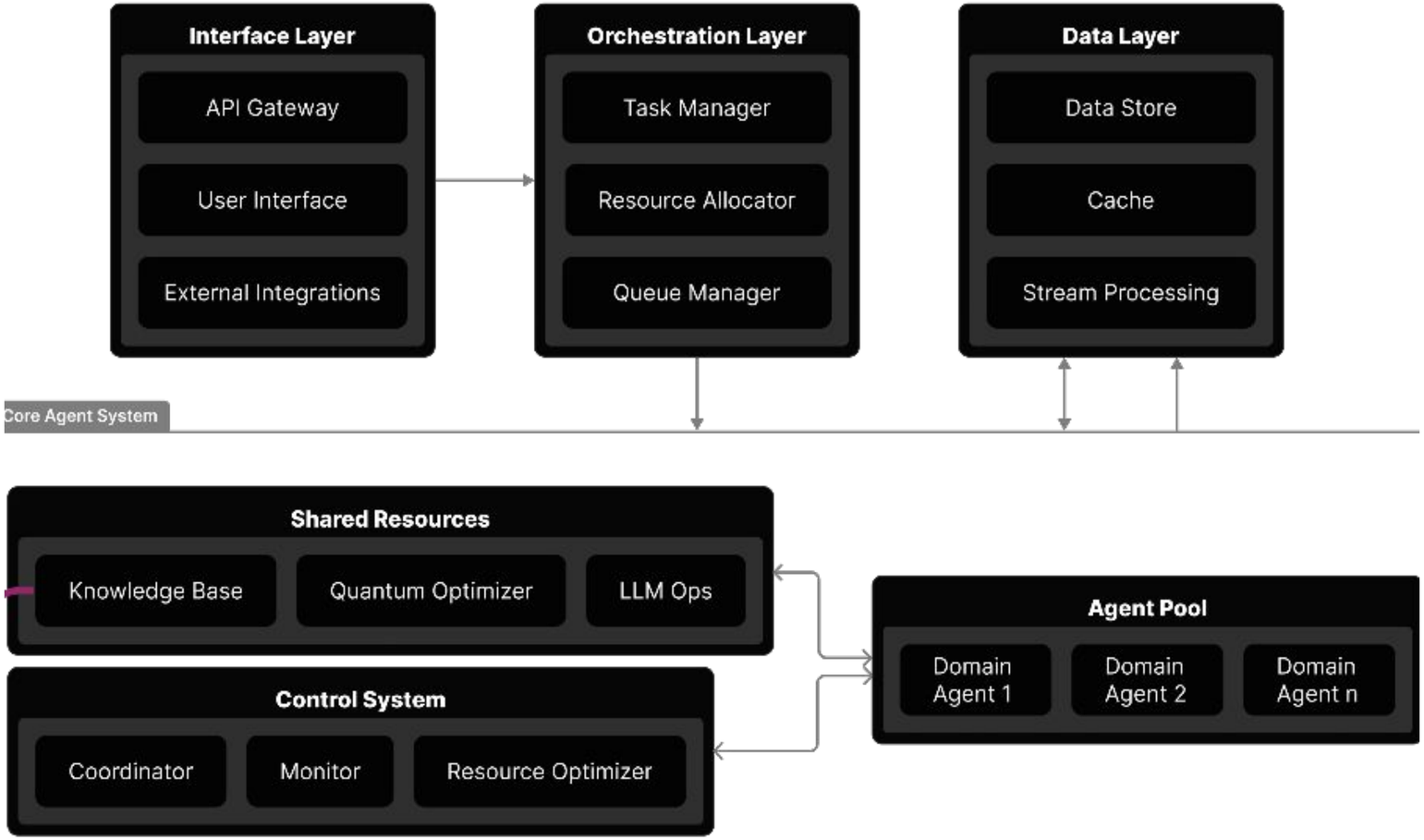


GUI based Multi Agent Systems Platform

QpiAI Agent Hive – Core Architecture

- Build Domain Targeted Multi Agent Systems from a pool of domain specialised agents.
  Drive 10 - 50x cost reduction with SLM based targeted agents.

- Interface with real-time streaming data, aggregate task focused logging and distributed tracing with best in class tools integration.
  Orchestrate across tools use, drive 10x efficiency gains.



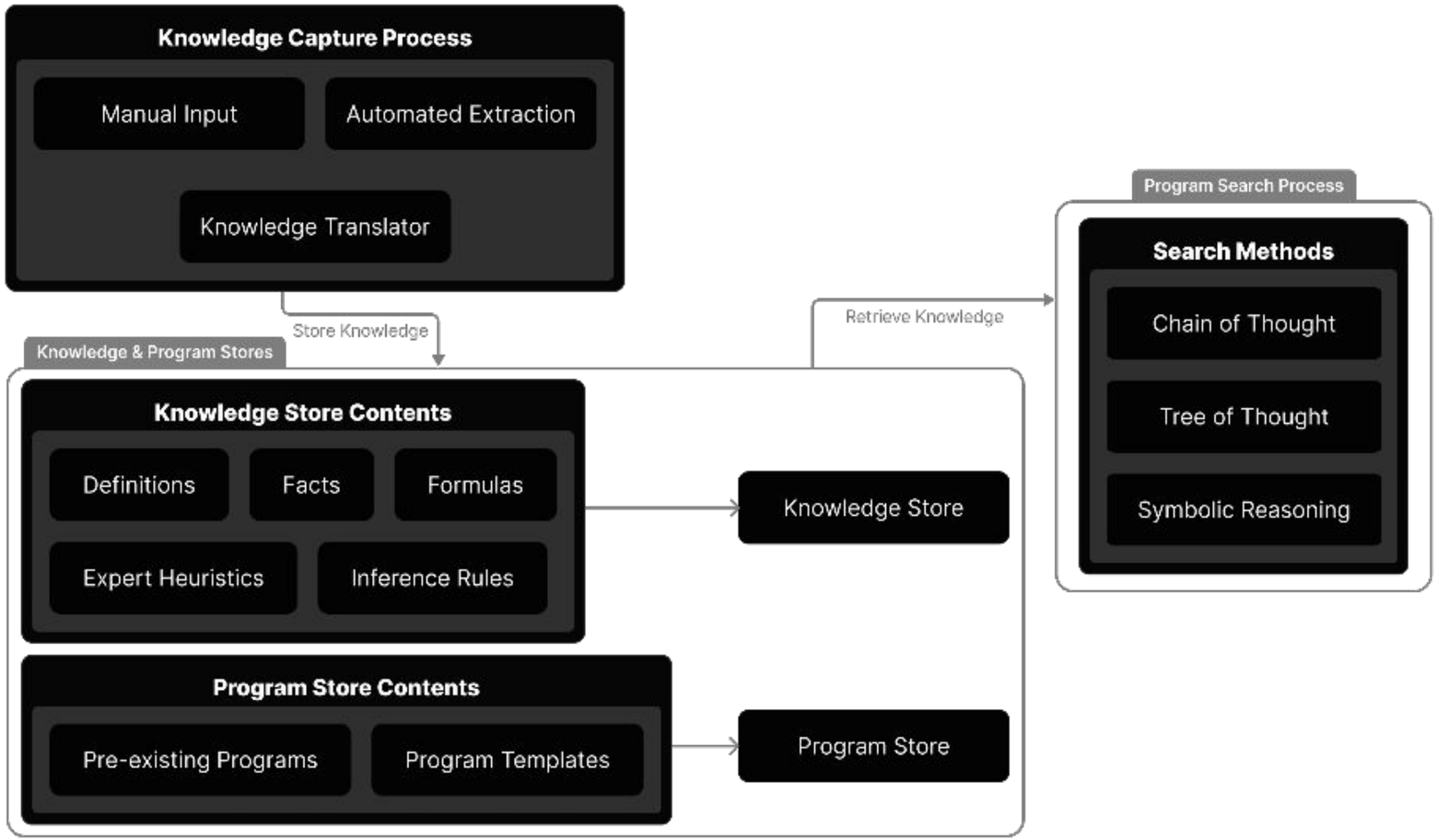
Dynamic Memory Adaptation with SLM Swarms

QpiAI Agent Hive – Knowledge Assimilation

The QpiAI Agent Hive Advantage

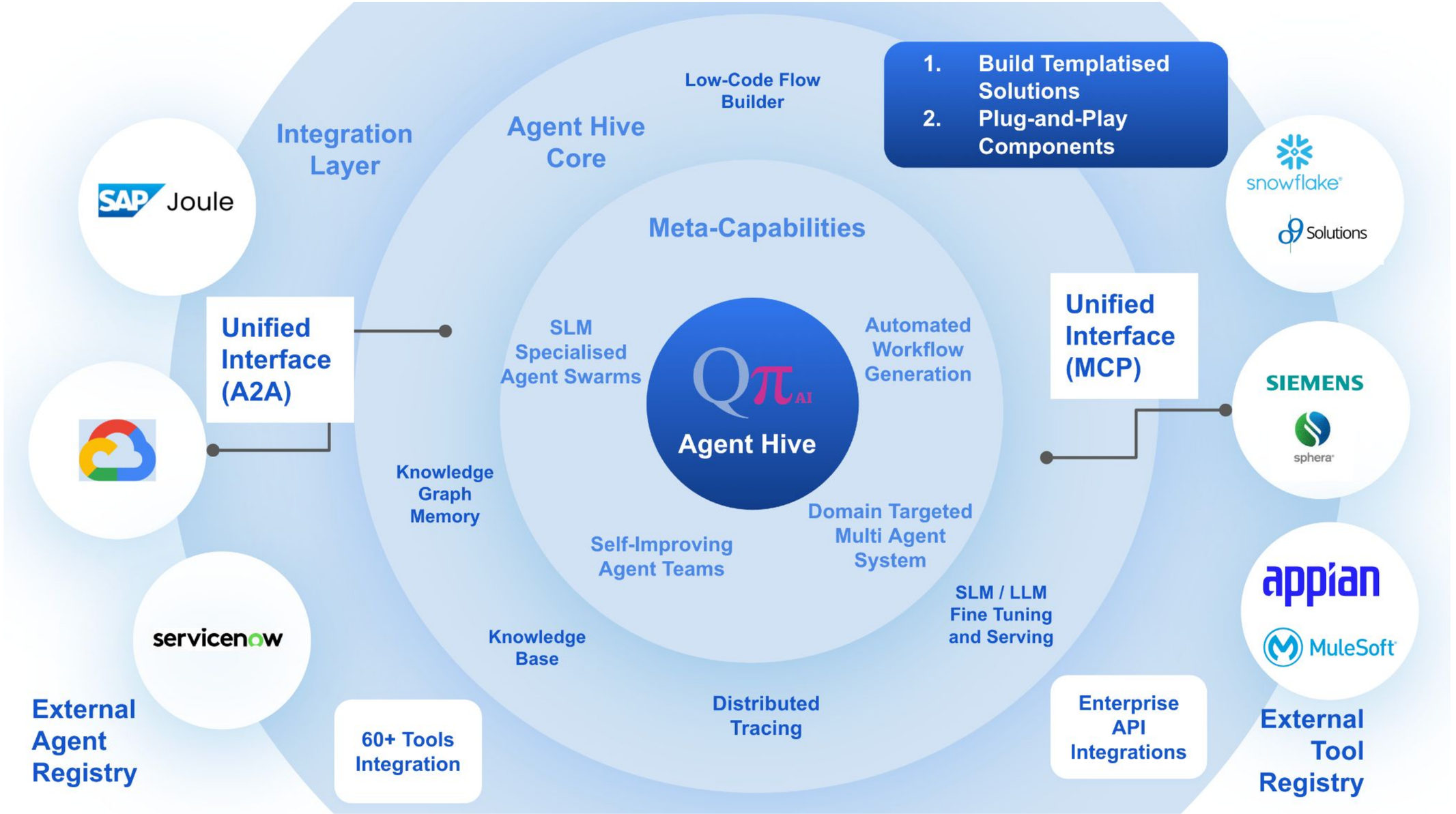
Reduce Hallucinations with:

- Effective Knowledge Capture from **Field Operator's Manual Inputs** to **Code Repositories** and Structured Knowledge Sources.
- Agentic Guardrails to pre-empt failure scenarios, integrate Human-in-the-loop feedback.



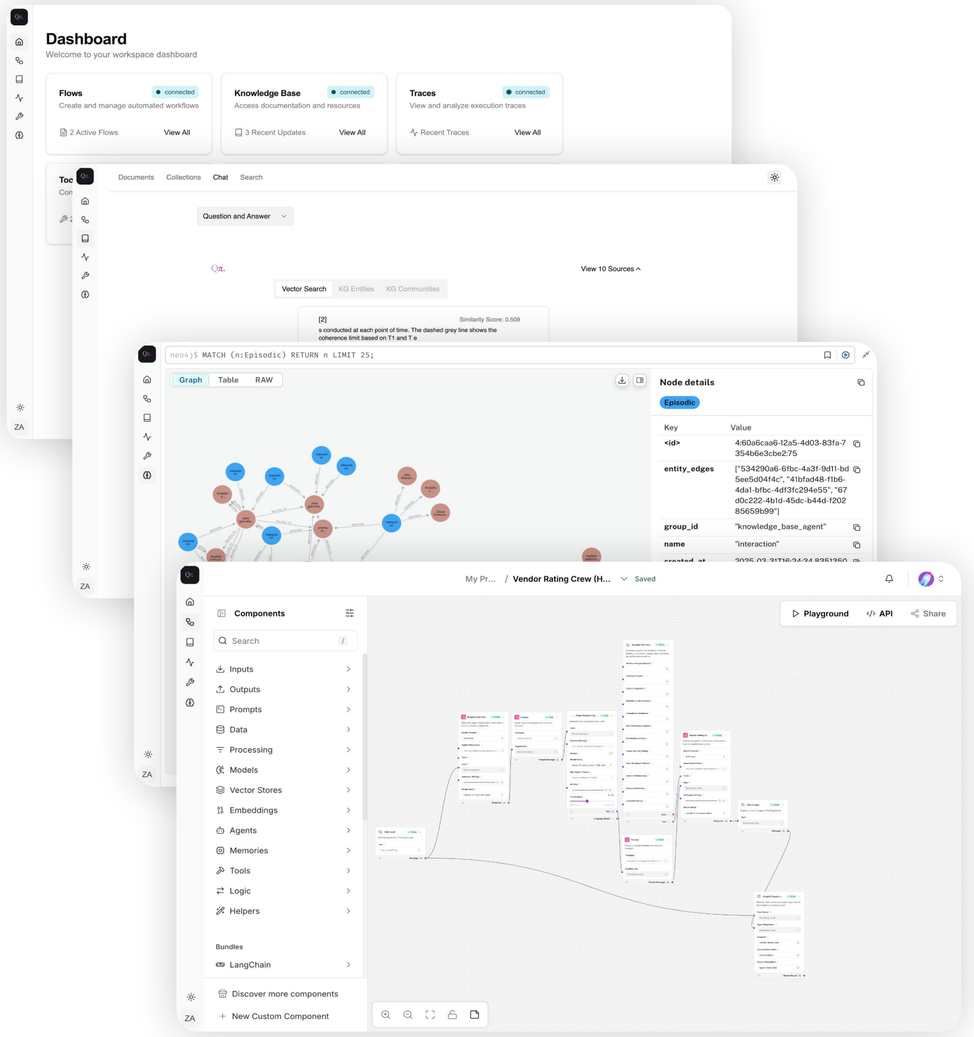
Rich Knowledge Assimilation Capabilities

QpiAI Agent Hive – Comprehensive Tools Integration



QpiAI Agent Hive – Core Feature List

- **GUI based Low Code** Multi Agentic Flow Builder
- **70+ integrated tools and MCP integration** for custom tools
- **Voice Agent** Functionality from get go
- Multi-Agent Workflow based **Observability** and **Distributed Tracing**
- **Dynamic Graph Memory** for Agents
- **Agentic Co-Pilot** For Industrial Automation
- **Agentic RAG** and Knowledge base ingestion
- Multi Agent **Orchestration**
- Enterprise **Deep Research Agent**
- Fully Managed **Cloud/On-Prem** Offering



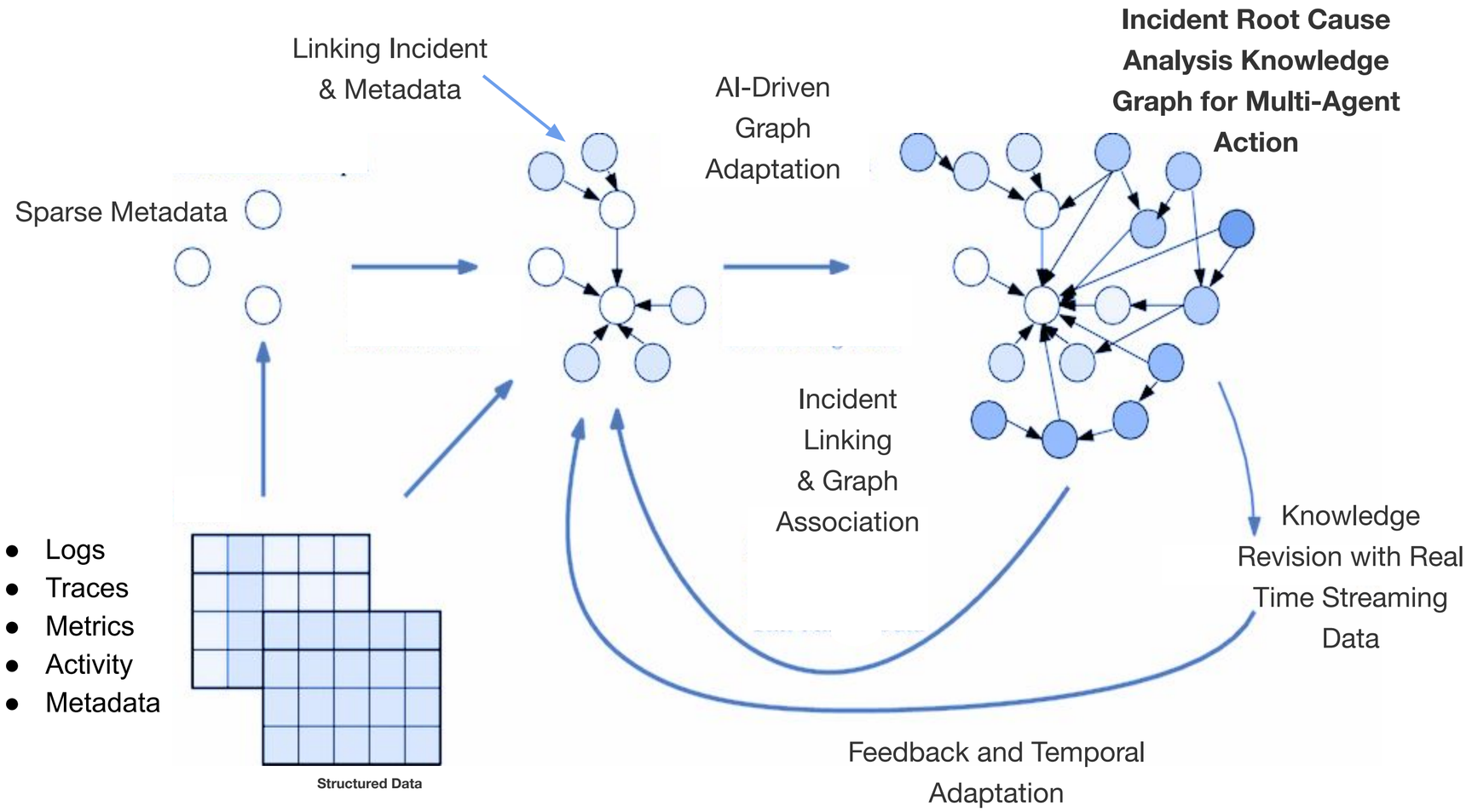
QpiAI Agent Hive – Competitive Analysis

Criteria	QpiAI Agent Hive	Amazon Bedrock Agents	Azure AI Foundry Agent	Vertex AI Agent Builder
Ease of Use	High: Modular, developer-friendly interface, suitable for technical and non-technical teams.	Moderate: Requires AWS expertise, steep learning curve for newcomers.	Moderate: Complex for non-Microsoft users, enterprise-focused setup.	High: Low-code interface, accessible for novices and developers.
Integration	Excellent: Flexible API and enterprise system integration with MCP and A2A, neutral platform across hyperscalers.	Strong: Seamless with AWS services (Lambda, S3), limited outside AWS.	Strong: Integrates with Microsoft tools (SharePoint, Azure Functions).	Strong: Leverages Google Cloud (BigQuery, Cloud Run), GCP-focused.
Multi-Agent Capabilities	Advanced: Domain-specific multi-agent systems with dynamic memory adaptation with SLM fine tuning.	Advanced: Supervisor-managed multi-agent collaboration.	Strong: Supports multi-agent workflows with orchestration.	Moderate: Emerging multi-agent features via ADK, in preview.
Scalability	Excellent: Scales for enterprises, rapid deployment across multi-tenant, on-prem and enterprise VPC.	Excellent: Highly scalable within AWS ecosystem.	Excellent: Enterprise-grade scalability in Azure.	High: Scalable via Google Cloud, rapid deployment focus.
Security & Compliance	Strong: Aligns with Global and Indian regulations, adaptable enterprise-grade security and compliance with hyperscalers, RBAC, custom guardrails.	Excellent: AWS Guardrails, global compliance standards.	Excellent: Microsoft Entra, RBAC, content filters.	Strong: RBAC, VPC controls, Google Cloud compliance.
Cost-Effectiveness	High: Competitive pricing for Global and Indian businesses, predictable costs.	Moderate: Pay-as-you-go, potentially high for non-AWS users.	Moderate: High costs for non-Microsoft users, enterprise pricing.	Moderate: Pay-as-you-go, cost-effective for GCP users.
Regional Support	Excellent: Local team, responsive support, separate support for India-specific solutions.	Moderate: Global support, limited regional customization.	Moderate: Global support, enterprise-focused.	Moderate: Global support, less focus on Indian market.

QpiAI Agent Hive – Use case in Industrial Automation for Diagnosis and Resolution of Real-Time Vehicle Incidents

The QpiAI Agent Hive advantage

- Root Cause Analysis Graph continuously adapted as new incidents emerge with alerts from Real-Time Streaming Data.
- Multi Agents reason with temporally adapted Knowledge Graph, generating hypothesis, ruling out conditions & narrowing down on the root cause.
- Graph adaptation with emerging trajectories of deteriorating health markers. Leverage unstructured meta-data to enrich context.



QpiAI Agent Hive – Intelligence layer for High Tech, High End Manufacturing

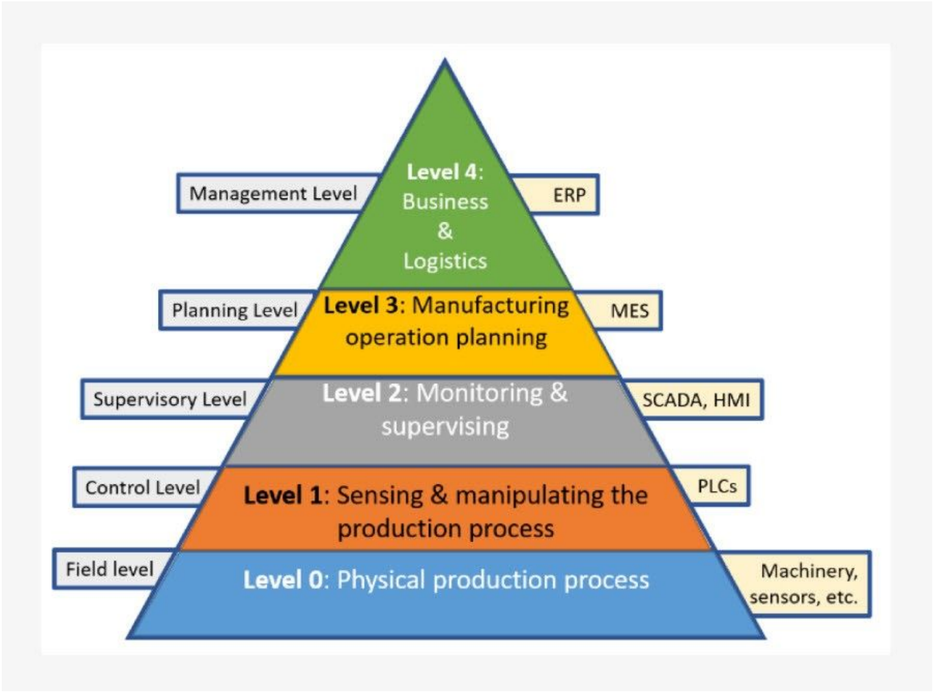
What:
Agent Hive as the **AI intelligence layer** between **MES** and **ERP.**

Why:

- Static pre-configured rules, limited flexibility for sudden changes / supply chain disruption / demand fluctuations.
- Legacy MES-ERP integrations are often rigid and can't keep pace with interconnected data stream complexity and scale.
- Cross System Orchestration: Optimize production schedules, manage supply chain risks, and self-correct issues.

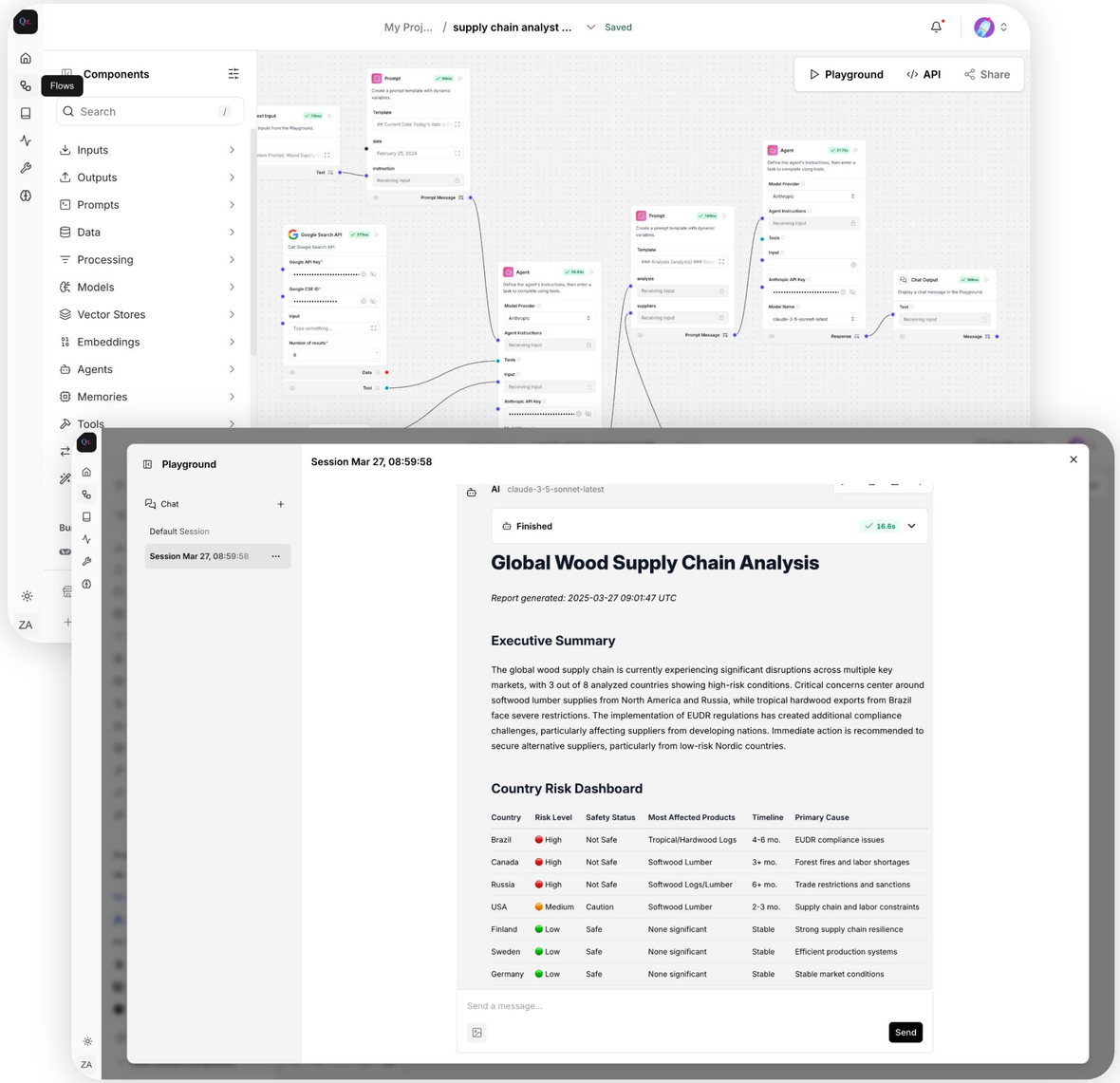
Focus Areas:

- Target Verticals: Semiconductor, Automotive, Aerospace, Battery Manufacturing and Pharma.
- Leverage partnerships between SAP and manufacturing majors like Honeywell, Rockwell etc.



The 5 Tier Automation Hierarchy

QpiAI Agent Hive – Agentic Supply Chain Resilience

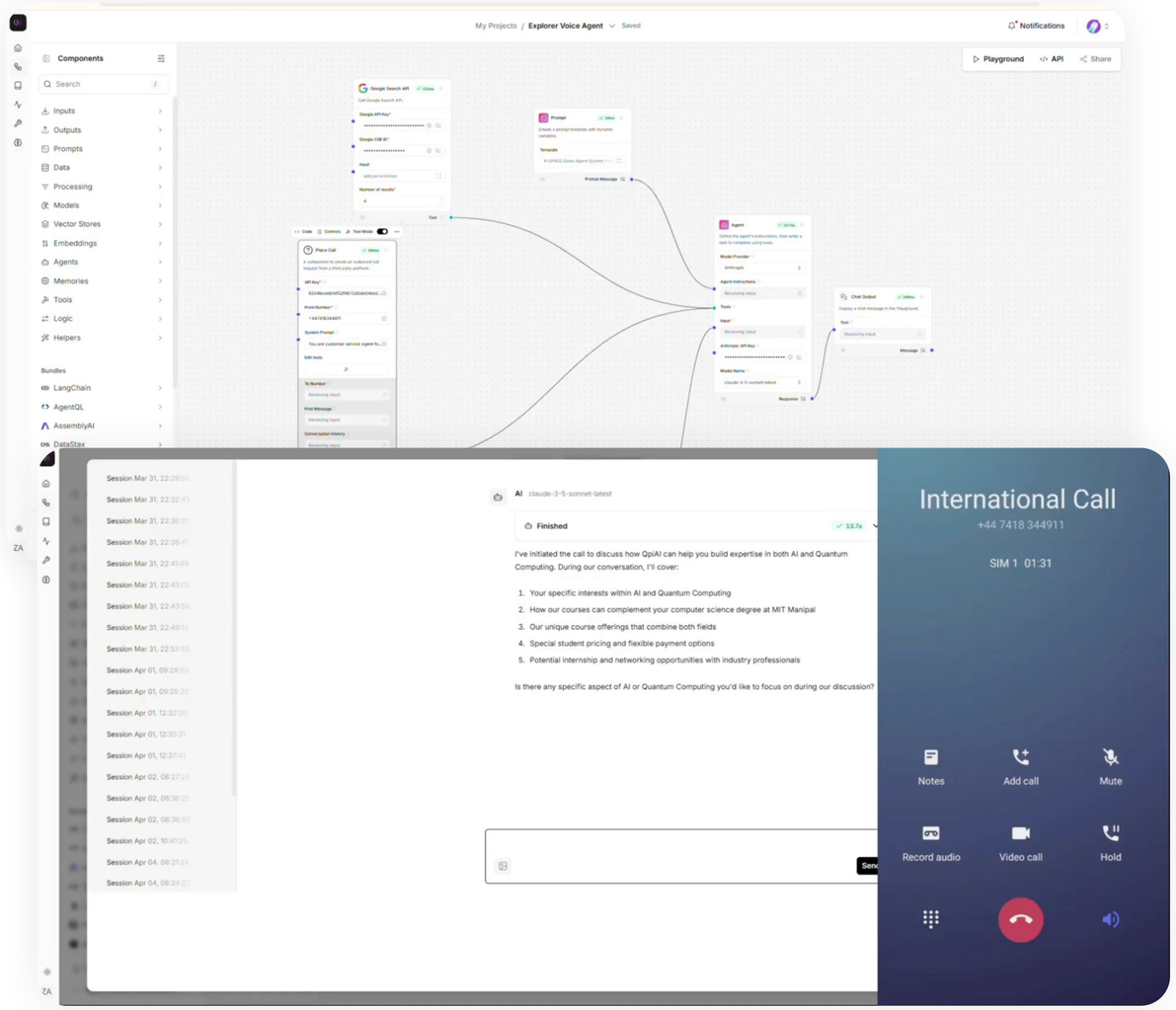


Supply Chain Analyst Crew

A Multi Agent System curating and analyzing Global News that could impact Raw Material Supply Chain for a Global Furniture Major in Europe, identify strategic preventive actions and recommend vendors for early negotiations to de-risk the supply chain.

With Agent Hive	Without Agent Hive
Time to first working prototype: **~3 hours**	Time to first working prototype: **3-5 days**
Unified Tooling for agents, tools, evals, memory	**Fragmented** Tooling
Production ready deployment from **first PoC**	Face Scaling Challenges after core features are proven
40x Cost Reduction with **Small Language Model Agents**	**Expensive Solution** using hosted Model Providers / **Overhead Fine Tuning** and **Maintaining Private Models**

QpiAI Agent Hive - Automated CX with Deep Research & Long Context Memory



Sales Support Voice Agent

A Sales Support Agent with access to tools, Deep Research and Voice Call - TTS, ASR, and VoIP all baked in.

With Agent Hive	Without Agent Hive
Time to first working prototype: **~1 hour**	Time to first working prototype: **3 days**
Unified Tooling for agents, tools and voice calls	**Fragmented** Tooling for TTS, ASR, and VoIP
Production ready deployment from **first PoC**	Face Scaling Challenges after core features are proven
25x Cost Reduction with **Small Language Model Agents**	**Expensive Solution** using hosted Model Providers / **Overhead Fine Tuning** and **Maintaining Private Models**

Use Cases with Manufacturing & Automotive Industry

Use Case	Problem Addressed	Key AI Agents	Initial Pilot Impact
Supply Chain Resilience and Risk Management	Supply chain disruptions (e.g., raw material shortages, logistics delays) require real-time, adaptive solutions; current systems lack dynamic risk management.	• Risk Monitoring Agent • Scenario Planning Agent • Mitigation Agent • Integration Agent	• **25%** improved risk prediction accuracy • **20%** reduction in supply chain downtime • **10-15%** savings on emergency logistics costs
Predictive Maintenance	Unplanned equipment failures cause downtime and high maintenance costs	• Equipment Monitoring Agent • Failure Prediction Agent • Maintenance Scheduling Agent	• **18-25%** reduction in maintenance costs • **5-15%** increase in production line availability
Production Scheduling and Optimization	Inefficient production scheduling leads to bottlenecks and missed deadlines	• Production Line Optimization Agent • Resource Allocation Agent • Order Management Agent	• **10-15%** increase in production throughput • **10-15%** reduction in lead times
Manufacturing Process Optimization and Quality Control	Manual quality control and process optimization in discrete manufacturing lead to defects and inefficiencies; precision is critical.	• Quality Inspection Agent • Forecast Agent • Optimization Agent • Feedback Agent	• **30%** reduction in defect rates • **25%** less production downtime • **15%** savings on rework and scrap costs
Energy and Resource Optimization for Sustainability	Energy and resource-intensive processes need real-time optimization to meet sustainability goals; current systems lack multi-factor efficiency.	• Resource Monitoring Agent • Resource Optimization Agent • Compliance Agent • Reporting Agent	• **20%** improved resource allocation accuracy • **15-20%** reduction in energy consumption • Enhanced ESG ratings with verifiable compliance

Use Cases for Supply Chain Resilience

Use Case	Problem Addressed	Key AI Agents	Initial Pilot Impact
Supplier Selection & Negotiation	Manual supplier selection is time-consuming and leads to suboptimal cost-quality trade-offs	• Supplier Profiling Agent • Market Analysis Agent • Negotiation Agent • Decision Agent	• **25%** reduction in suboptimal selections • **40%** faster procurement cycles • **10-15%** cost savings
Real-Time Inventory & Demand Synchronization	Misaligned inventory levels due to fluctuating demand patterns	• Demand Forecasting Agent • Inventory Agent • Procurement Agent • Logistics Agent	• **20-30%** improved forecast accuracy • **15%** reduction in stockouts • **10-20%** lower inventory costs
Sustainability-Driven Supplier Compliance	Difficulty ensuring suppliers meet sustainability standards and certifications	• Compliance Agent • Risk Assessment Agent • Sourcing Agent • Reporting Agent	• **30%** fewer compliance errors • **100%** sustainable sourcing visibility • **60%** less audit time
Risk Mitigation in Global Supply Chains	Vulnerability to disruptions like port delays, tariffs, and natural disasters	• Risk Monitoring Agent • Scenario Analysis Agent • Mitigation Agent • Communication Agent	• **25%** better risk prediction • **20%** less supply chain downtime • **10-15%** savings on emergency costs
Automated Contract Management	Labor-intensive contract administration and inconsistent performance tracking	• Contract Agent • Performance Agent • Analytics Agent • Renewal Analyst Agent	• **20%** fewer contract errors • **50%** less administrative time • **35%** improved supplier performance

Vertical Focused Use Cases with SAP

Use Case	Problem Addressed	Key Agents	Potential Improvements
End-to-End Procurement Process Optimization	Complex procurement workflows require context preservation across multi-step processes; reducing delays and errors.	• Supplier Profiling Agent • Process Optimization Agent • Negotiation Agent • Workflow Agent	• **25%** reduction in supplier selection errors • **40%** faster procurement cycles • **10-15%** cost savings
Supply Chain Resilience and Risk Management	Supply chain disruptions (e.g. raw material shortages, logistics delays) require real-time, adaptive solutions; improving dynamic risk management.	• Risk Monitoring Agent • Scenario Planning Agent • Mitigation Agent • Integration Agent	• **25%** improved risk prediction accuracy • **20%** reduction in supply chain downtime • **10-15%** savings on emergency logistics costs
Manufacturing Process Optimization and Quality Control	Manual quality control and process optimization in discrete manufacturing lead to defects and inefficiencies; precision is critical.	• Quality Inspection Agent • Forecast Agent • Optimization Agent • Feedback Agent	• **30%** reduction in defect rates • **25%** less production downtime • **15%** savings on rework and scrap costs
Energy and Resource Optimization for Sustainability	Energy and resource-intensive clients need real-time optimization to meet sustainability goals; adding multi-factor efficiency.	• Resource Monitoring Agent • Resource Optimization Agent • Compliance Agent • Reporting Agent	• **20%** improved resource allocation accuracy • **15-20%** reduction in energy consumption • Enhanced ESG ratings with verifiable compliance
Cross-Functional Workflow Automation with Joule Integration	Siloed cross-functional workflows (e.g. procurement, finance) - improve coordination for multi-step processes with interoperable context.	• Context Agent • Task Agent • Coordination Agent • Integration Agent	• **20%** fewer process errors • **40%** faster cross-functional workflows • **15-20%** improved cross-departmental productivity

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Technical Use Cases with SAP

Use Case	Description	Key Agents	Potential Improvements
Advanced Analytics for Supply Chain Optimization	Enhance SAP's supply chain management with real-time, multi-variable analytics for demand forecasting, inventory optimization, and risk mitigation, leveraging multi-agent systems to complement SAP Joule's capabilities.	• Supplier Profiling Agent • Logistics Optimization Agent • Inventory Management Agent • Demand Forecasting Agent	• **15-25%** cost savings • **20%** better delivery times • Low latency for near real-time decisions
Real-Time Predictive Maintenance for Manufacturing	Improve equipment uptime in manufacturing with low-latency, real-time predictive maintenance, integrating with SAP Manufacturing Execution Systems to enhance SAP Joule's operational insights.	• Equipment Monitoring Agent • Failure Prediction Agent • Maintenance Scheduling Agent	• **25%** less downtime • **30%** lower maintenance costs • **90%** accuracy in sensor data processing
ESG Compliance Automation for Oil & Gas	Automate ESG reporting and compliance checks for oil & gas, integrating with SAP Sustainability Control Tower, leveraging multi-agent systems to enhance SAP Joule's emerging ESG features.	• Compliance Agent • Reporting Agent • Resource Monitoring Agent • Resource Optimization Agent	• **95%** compliance accuracy • **40%** faster audits • **15-20%** reduction in energy consumption
Custom Workflow Orchestration for Cross-Functional Processes	Enable complex, cross-functional workflows across SAP modules, enhancing SAP Build with low-code/no-code tools, complementing SAP Joule's capabilities.	• Context Agent • Task Agent • Coordination Agent • Integration Agent	• **30%** faster setup • **20%** fewer process errors • **95%** uptime for cross-functional tasks
Non-SAP System Integration for Hybrid Environments	Extend AI capabilities to non-SAP systems for enterprises with hybrid IT landscapes - on premise and cloud, enhancing SAP Data Sphere integration, complementing SAP Joule's cloud focus.	• Integration Agent • Context Agent • Task Agent • Coordination Agent	• **25%** faster deployment • **20%** better integration efficiency • **85%** accuracy in cross-system workflows

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Use Cases with Oil & Gas Industry

Use Case	Problem Addressed	Key AI Agents	Initial Pilot Impact
Supply Chain Risk Mitigation and Logistics Optimization	Supply chain disruptions from geopolitical risks, port delays, and natural disasters cause costly downtime and emergency logistics expenses.	• Risk Monitoring Agent • Scenario Agent • Mitigation Analysis Agent • Logistics Management Agent	• **25%** improved risk prediction accuracy • **20%** reduction in supply chain downtime • **10-15%** savings on emergency logistics costs
Energy Optimization and Emissions Reduction	Pressure to reduce energy consumption and carbon emissions to meet ESG goals and regulations, hindered by lack of real-time, multi-factor optimization.	• Resource Monitoring Agent • Resource Optimization Agent • Compliance Agent • Reporting Agent	• **20%** improved energy allocation accuracy • **15-20%** reduction in energy consumption and emissions • Enhanced ESG ratings with verifiable compliance
Drilling and Exploration Optimization	Complex upstream decisions about site selection, drilling parameters, and resource allocation lead to high costs and environmental risks.	• Geodata Analysis Agent • Planning Agent • Risk Assessment Agent • Execution Analysis Agent	• **20%** improved drilling site selection accuracy • **15%** reduction in drilling costs • **25%** lower operational risks
Refinery Process Optimization and Predictive Maintenance	Energy-intensive refining processes and equipment failures cause downtime, high maintenance costs, and inefficiencies.	• Process Monitoring Agent • Process Optimization Agent • Predictive Maintenance Agent	• **20%** improved process efficiency • **25%** reduction in downtime • **15%** savings on maintenance and energy costs
Procurement and Supplier Management Automation	Manual procurement processes for complex supplier networks lead to suboptimal sourcing, delays, and compliance issues.	• Supplier Profiling Agent • Negotiation Agent • Compliance Agent • Workflow Agent	• **25%** reduction in supplier selection errors • **40%** faster procurement cycles • **10-15%** cost savings through optimized sourcing



Thank You!

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www.qpiai.tech

Appendix: Technical Competitive Analysis

Criteria	QpiAI Agent Hive	Amazon Bedrock Agents	Azure AI Foundry Agent	Vertex AI Agent Builder
Core Technology	Dynamic Memory Adaptation with SLM Swarms: Adaptive context retention, domain targeted & cost-effective.	Natural Language-Driven Logic Checking: Reduce hallucinations.	Unified Orchestration Layer: Real-time telemetry, governance.	Grounding in Enterprise Data: Accurate, context-aware responses.
Architecture	Modular, event-driven, GUI based workflows, decentralized orchestration.	Serverless, API-driven, centralized orchestration.	Cloud-native, microservices, centralized orchestration.	Hybrid, no-code/code-driven, GCP-integrated.
Model Support	Open Source SLMs/LLMs, custom deployment. GUI based Distributed tracing for distillation & fine-tuning.	Third-party FMs (Claude, Llama), RAG, SageMaker for fine-tuning.	GPT-4o, Llama, fine-tuning, distillation.	Gemini, AutoML, LangChain/LlamaIndex integration.
Tool Integration	Flexible API and enterprise system integration with MCP and A2A.	AWS services (Lambda, S3), custom APIs, AWS-specific configs.	Microsoft tools (Bing, SharePoint), Azure Functions, APIs.	Google Search, RAG Engine, APIs, webhooks.
Multi-Agent Orchestration	Decentralized Coordination Engine: Autonomous, scalable collaboration.	Supervisor-agent model: Centralized, may introduce latency.	Centralized orchestration: Structured logging, retries.	Emerging ADK (preview), A2A protocol for interoperability.
Security	E2E encryption, RBAC, DPDP Act compliance, India data residency.	Bedrock Guardrails, VPC, IAM, global compliance (GDPR, HIPAA).	Microsoft Entra, RBAC, content filters, global/regional compliance.	RBAC, VPC controls, HIPAA/ISO compliance, data grounding.
Indian Market Fit	Excellent: India-based, DPDP Act compliance, vernacular use cases, local support.	Moderate: Global focus, less tailored for Indian Enterprises.	Moderate: Enterprise-focused, complex for startups.	Moderate: Global platform, limited regional customization.